

AutoCert using Custom Studios and Actions TOI

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Last Update: 28.5.2026 V1.1.2

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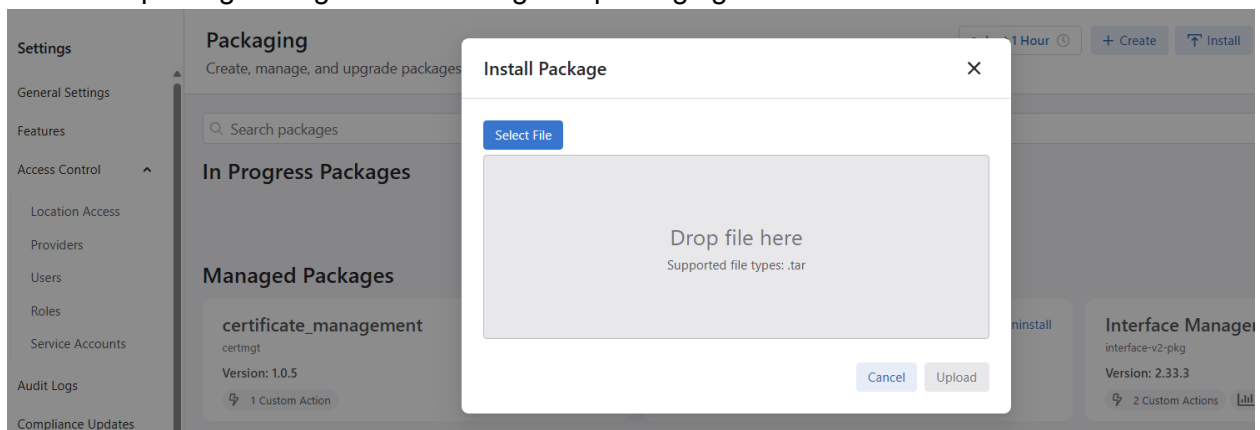
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Prerequisites:

1. Download certificates to the local directory on your laptop.
2. Importing custom studio actions and certificates and creating an action bundle
3. For new switches where CVP is already configured with prerequisite, you can skip directly to section: Applying the Action Bundle.
4. Copying the certificates from the EST/RadSec server can be done manually using many ways such as scp/ssh/terminal.
5. If you already uploaded the certificates and created a self-signed certificate you can skip directly to section: "Configure the Studio"

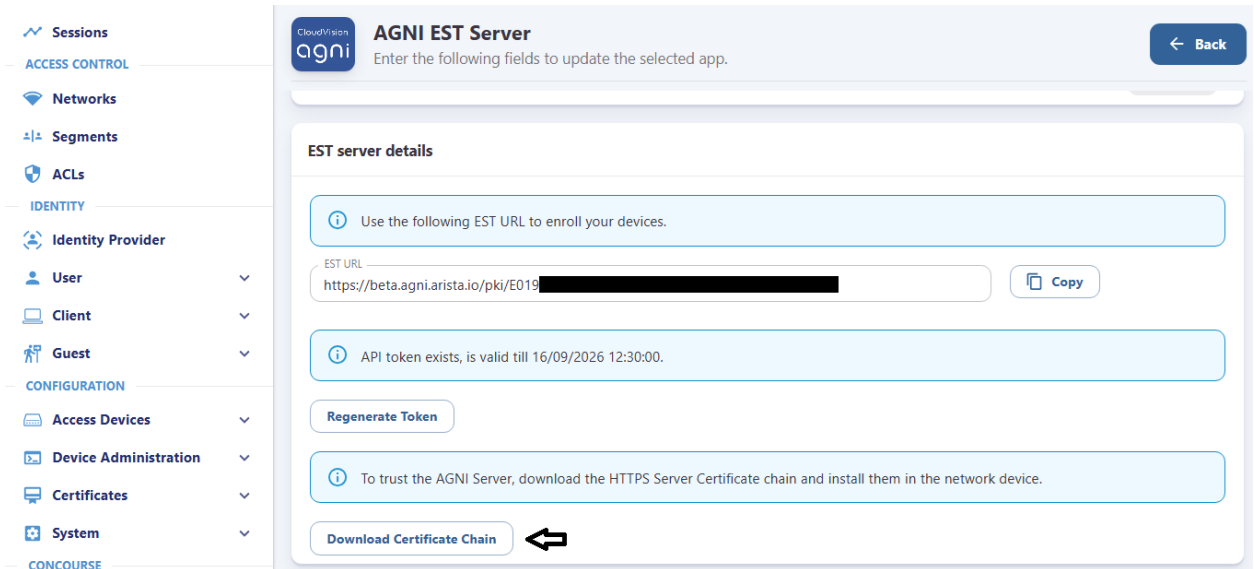
Download and import the custom studio and actions

6. Download the package, this includes both custom actions and the custom studio.
[pkg-est-certmgt_x.x.x.tar](#)
7. Install the package using CVP -> settings -> packaging -> install



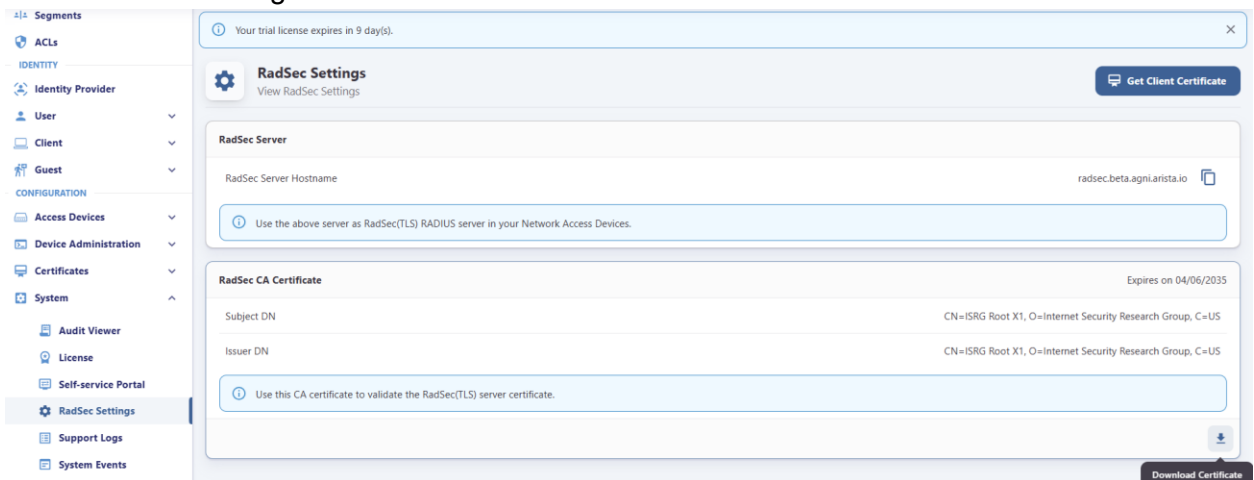
Download The EST Root CA from AGNI

1. Go to AGNI -> Concourse, install the AGNI EST application and download the EST CA certificate.



Download The RadSec Root CA from AGNI

1. Go to AGNI -> setting and download the EST CA certificate



Create an Action Bundle

1. The Action Bundle will be used to create a self-signed certificate and for each switch. It will also push both Radsec and EST CAs into the switches certificate directory
2. The Action Bundle should include 4 actions:
 - 2.1. Set configuration
 - 2.2. Create self signed certificate
 - 2.3. Push certificate with the Radsec CA

- 2.4. Push Certificate with the EST CA
- 3. Example of creating the bundle:
 - 3.1. CVP -> Provisioning -> Action Bundles -> new Action Bundle

New Action Bundle

Bundle Name

Bundle - Add Root CAs

Description (optional)

This bundle will create a self sign certificate and push needed CAs to switch

Add action... ▼

Series

Parallel

- 3.2. Click on the pencil sign to edit the newly created bundle

☐	Name ↕	Description ↕	Actions
☐	Bundle - Add Root CAs	This bundle will create a self sign certificate and push needed CAs to switch	—  

- 3.3. Add action bundle (1) - set configuration

Action Bundle: Bundle - Add Root CAs

Bundle Name

Bundle - Add Root CAs

Description (optional)

This bundle will create a self sign certificate and push needed CAs to switch

Set Configuration ▼

⬇ Series

⊞ Parallel

1. Set Configuration



Push a designed config or roll back to a previous running config. This is accomplished by providing a DeviceID, Timestamp, and Source (Running Config or Designed Config).

DeviceID ⓘ

- 3.4. Add action bundle (2) - Self-signed certificate/key creation vx.y.z.
 - 3.4.1. Copy both file names of the certificate and key, we will need to use it later in the custom studio details.
 - 3.4.2. deviceId select: "provide via template"
 - 3.4.3. Make sure to change the default values as needed

2. Self-signed certificate/key creation v1.0.5



This action will generate a self-signed certificate and an associate key on the devices that can be used in an SSL profile.

DeviceID ⓘ

Select a device...



cert_file ⓘ

self-signed.crt

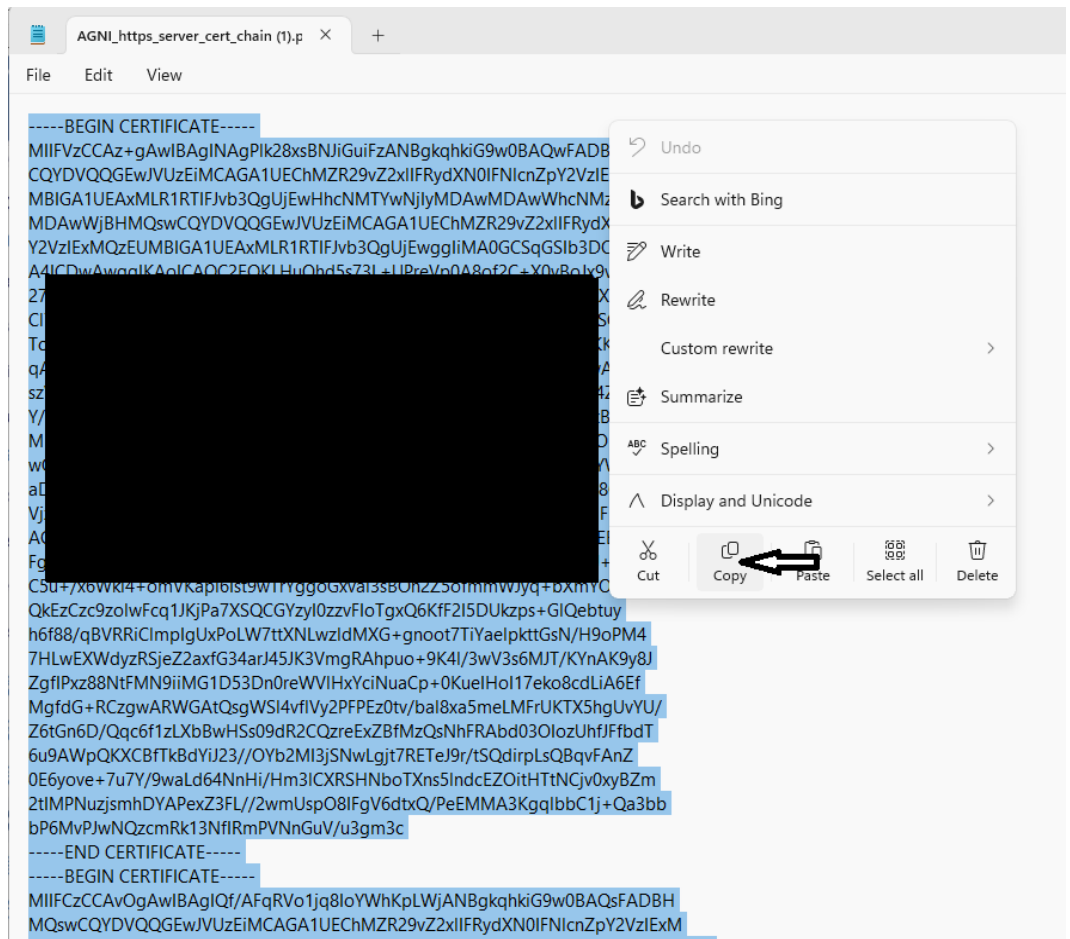
country ⓘ

IL

email ⓘ

test@test.com

- 3.5. Add action bundle (3) - Push CA Certificate to switch vX.Y.Z
 - 3.5.1. This action will be used to push the EST certificate into the switch.
 - 3.5.2. Certificate must be in PEM format.
 - 3.5.3. device ID select: "provide via template".
 - 3.5.4. Filename: temporary location for the certificate file.
*To avoid syntax errors, it is recommended to use the certificate filename originally received from AGNI for the Radec default is: "AGNI_https_server_cert_chain.pem"
 - 3.5.5. Open the downloaded EST certificate chain using notepad and copy it as one line. The script will format it back with newlines.



3. Push CA Certificate to switch v1.0.6



Downloads and copy a certificate (or chain) into the eos' certificate directory

DeviceID ⓘ

Provide via template



certificate ⓘ

-----BEGIN CERTIFICATE----- MIIFVzCCAz+gAwIBAgINAglk28xsBNJ

filename ⓘ

/tmp/AGNI_https_server_cert_chain.pem

- 3.6. Add action bundle (4) - Push CA Certificate to switch vX.Y.Z
 - 3.6.1. This action will be used to push the RadSec certificate into the switch.
 - 3.6.2. Certificate must be in PEM format.
*It is the same action and process as the latest, the only difference is the certificate and filename.
 - 3.6.3. device ID select: "provide via template".
 - 3.6.4. Filename: temporary location for the certificate file.
*To avoid syntax errors, it is recommended to use the certificate filename originally received from AGNI for the Radec default is: "radsec_ca_certificate.pem"
 - 3.6.5. Open the downloaded RadSec certificate chain using notepad and copy it as one line. The script will format it back with newlines.

4. Push CA Certificate to switch v1.0.6

↑ · ↓ · 🗑️

Downloads and copy a certificate (or chain) into the eos' certificate directory

DeviceID ⓘ

Provide via template ▼

certificate ⓘ

-----BEGIN CERTIFICATE----- MIIFazCCA1OgAwIBAgIRAIQz7DSQON

filename ⓘ

/tmp/radsec_ca_certificate.pem

3.7 Click Save to save the bundle

3.8 Create a new Template and add the bundle into the temple

Provisioning

Studios

Tags

Workspaces

Change Control

Action Bundles

Templates

Actions

Zero Touch Provisioning

Templates

Configure tem

Search Temp

1 Item

☐ Name

☐ push trus

New Template

✕

* Name

Push certificated to switches

Description

Cancel

Save

Live Time

shaip

internal...ista.co

+ New Template

🔗 🗑️

Push certificated to switches

Enter template description... 

 Parallel  Stage Rules

Unsaved Changes

Save Template

 Stage Rule Name 

 Move Down  Move Up  Delete  Add Stage Rule

Action Bundle  [+ Create Bundle](#)

Device Filter 

Arrange Bundles  Series 

Applying the Action Bundle

1. ** Note that the action will overwrite any existing certificates using the same filename including any self signed certificates already existing on the switch. Make sure the filenames are not in use.
2. CVP -> provisioning -> change control -> create change control
3. Change the name for example: “push certificate to new switches”
4. add action -> “Set Configuration”
5. choose the devices you want to push onto the certificates. Make sure you only choose the desired devices as this will override existing configuration.
6. Click “Add to change control”.
You should now see the devices listed on the change control.
7. Click “Apply Template” and select the template created.
You should now see 4 actions per device.
Click “Review and Approve” to apply the config.

Pushing Certificates to new switches

Name

Pushing Certificates to new... 

Description

-- 

Schedule Start

Select date



Search actions

Select a Template 

Change Control Stages (Empty) 

Start Building Your Change

To get started, let's add some Actions

+ Add Action

Pushing Certificates to new switches

Name

Pushing Certificates to new... 

Description

-- 

Schedule Start

Select date



Search actions

Select a Template 

Change Control Stages (Empty) 

Start Building Your Change

To get started, let's add some Actions

+ Add Action

← Add Action

Action ⓘ

Set Configuration

Config Source

Designed Config

Run action against selected devices

☒ Filter devices

☐ C200-0923

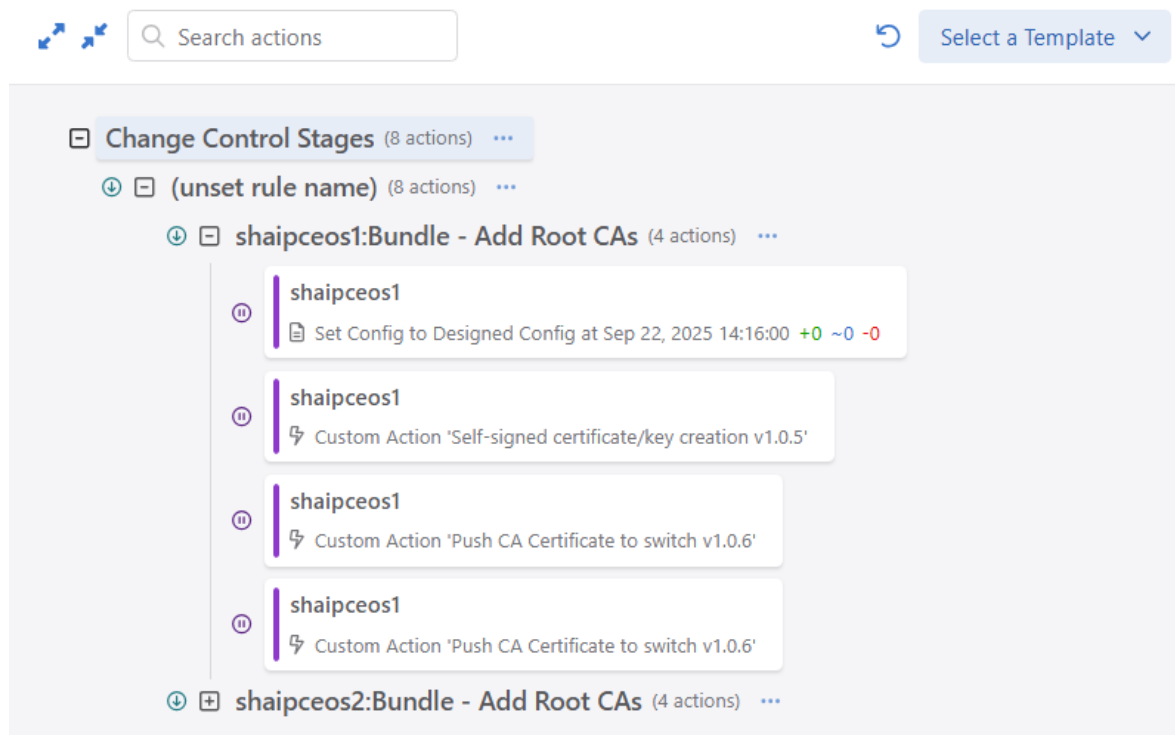
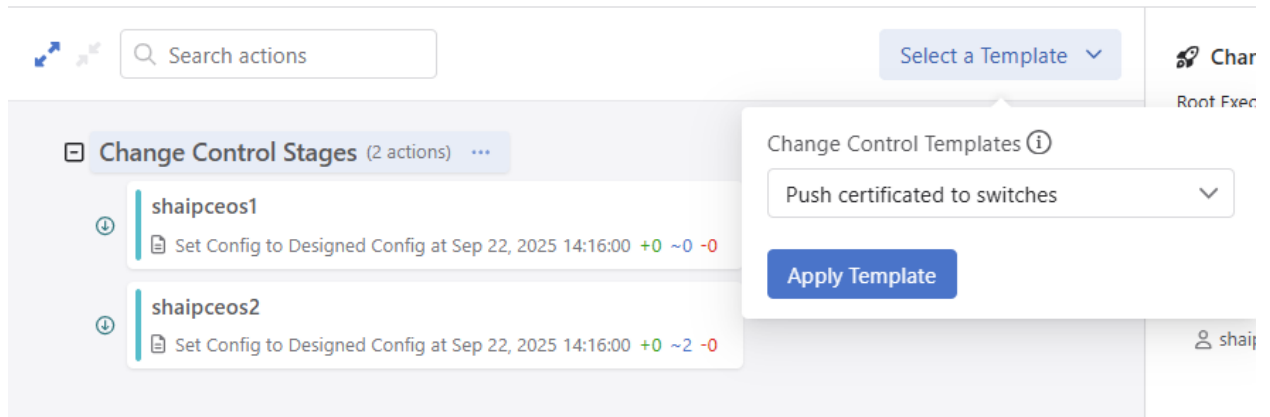
Inactive

☒ shaipceos1

4.34.1F | Inactive | Provisioned

☒ shaipceos2

4.34.1F | Inactive | Provisioned



Configure The Studio

1. CVP -> provisioning -> studios -> "Configure AGNI EST vX.Y.Z"
2. Create a new workspace.
3. Device selection: select the devices you want to apply the configuration.
You can leave it as "all devices" and filter the exact devices by adding them inside the studio.
4. Apply devices: choose the devices you want to manage.
 - 4.1 All devices using the same configuration should be in the same line.

- 4.2 devices using different server details, or different certificate properties use separated lines.
5. Add General dot1x/radius config:
 - 5.1 Yes - If you want the studio to build basic radius and dot1x configuration. This configuration will use the auto generated certificate for the Radsec.
 - 5.1 No - For more advanced radius/dot1x configuration please use “authentication” and “campus L3” built-in studios
 6. **certificate properties:** Click “view” to enter the configuration.
 - 6.1. Make sure you edit all the configurations according to your company policies.
 - 6.2. **Common-name:** should be left blank so that the switch’s mac will be used. This is required for AGNI to first identify the device.
 - 6.3. **auto cert profile name:** the name of the profile will be the name of the certificate file automatically generated by EST. this certificate name can be used later for Radsec configuration.
 - 6.4. **self sign certificate** file name: use the filename pre-created on the switch.
 - 6.5. **self sign key file name:** use the filename pre-created on the switch.
 7. **EST server details:** Click “view” to enter the configuration.
 - 7.1. **est server url:** To see the est url go to AGNI -> CONCOURSE -> go to AGNI-> concourse -> EST server details for example:
<https://beta.agni.arista.io/pki/xxx-xxxx-xxxxxxx>
 - 7.2. **est ssl server profile name** name of the ssl profile to configure on the switch for example: agni-server
 - 7.3. **radius server hostname:** hostname of the radius server . To see the hostname go to AGNI -> System -> RadSec Settings for example: [radsec.beta.agni.arista.io](https://beta.agni.arista.io/radsec)
 - 7.4. **radius ssl server profile name:** the name of the ssl profile used to authenticate the radisec server certificate for example: agni-server-radsec
 - 7.5. **radius server group name:** the name of the ssl profile used to authenticate the radisec server certificate for example : agni-server-group
 - 7.6. **radius source interface:** optionally specify an interface for radius traffic originating from the switch.
 8. **Tokens:**
 - 8.1. AGNI EST Server -> click on re/generate Token.
 - 8.2. This token is shown only once on the EST server, if you regenerate the token you cannot use the previous token, so all switches will need to be updated with the new token.
 - 8.3. **enrollment token:** token used to authenticate enrollment process with the EST server
 - 8.4. **reenrollment token:** token used to authenticate reenrollment process with the EST server.

- 8.5. In the current AGNI EST implementation reenrollment token and reenrollment token are the same
- 8.6. You can copy the token as is, or copy already encrypted password by adding 7/8a "7xxxxxx" or "8a xxxx" for an already unfocused password.

Device Selection

This studio is assigned to:

device: *[Edit](#)

Apply devices

Apply devices	↓↑	Add general dot1x/radius config	↓↑	certificate properties ⓘ	est server details	Tokens	🔍
⋮ device: shaipceos2		Yes	▼	View >	View >	View >	🗑️
⋮ device: shaipceos1		Yes	▼	View >	View >	View >	🗑️
+ Add Apply device							

country

IL/US etc..

country ⓘ

state

IL/US etc

state ⓘ

common-name

leave blank if you want studio to use device
mac-address - recommended

common-name ⓘ

subject-alternative-name email

used for the certificate's SA URI for example:
shaip@arista.com

subject-alternative-name email ⓘ

subject-alternative-name uri

used for the certificate's SA URI for example:
<http://mysite.mydomain.com>

subject-alternative-name uri ⓘ

auto cert profile name

name of the profile for the auto certificate
profile, this will also be used as the name of
the certificate file for example: myautocert

auto cert profile name ⓘ

self sign certificate file name

self sign certificate file name

self sign key file name

self sign key file name

est server url

To see the est url go to AGNI ->
CONCOURSE -> go to AGNI-> concourse ->
EST server details for example:
`https://beta.agni.arista.io/pki/xxx-xxxx-
xxxxxxxx`

est server url ⓘ

est ssl server profile name

name of the ssl profile to configure on the
switch for example: agni-server

est ssl server profile name ⓘ

radius server hostname

hostname of the radius server . To see the
hostname go to AGNI -> System -> RadSec
Settings for example
`radsec.beta.agni.arista.io`

radius server hostname ⓘ

radius ssl server profile name

the name of the ssl profile used to
authenticate the radisec server certificate for
example: agni-server-radsec

radius ssl server profile name ⓘ

radius server group name

the name of the ssl profile used to
authenticate the radisec server certificate for
: agni-server-group

radius server group name ⓘ

radius source interface

radius source interface

> [Configure AGNI EST v1.0.7](#) / [devices:shaipceos2](#) / [Tokens](#)

enrollment token

enrollment token

reenrollment token

reenrollment token

Identity Provider
 User
 Client
 Guest
 CONFIGURATION
 Access Devices
 Device Administration
 Certificates
 System
 Audit Viewer
 License
 Self-service Portal
 RadSec Settings
 Support Logs
 System Events
 Notification Settings

Your trial license expires in 8 day(s).

AGNI EST Server
 Enter the following fields to update the selected app.

This app provides EST service to securely provision RadSec client certificates for network devices.

Cancel Update

EST server details

Use the following EST URL to enroll your devices.

EST URL

 Copy

API token exists, is valid till 16/09/2026 12:30:00.

Regenerate Token

Back

Example Generated configuration:

management security

auto-certificate protocol est agni-server

server url https://beta.agni.arista.io/pki/E019xxxxxxxx

server ssl profile agni-server

credentials enroll token 7 0603160B444C2E1A0C33xxxxx

credentials re-enroll token 7 0603160B444C2E1A0C381Bxxxxx

!

auto-certificate profile myautocert

key self-signed.key

parameters distinguished-name common-name 00:1c:73:f9:d1:19

parameters distinguished-name country IL

```

parameters distinguished-name state IL
parameters subject-alternative-name ip 172.20.20.4
parameters subject-alternative-name dns shaipceos1
parameters subject-alternative-name email shaip@arista.com
parameters subject-alternative-name uri http://shaip.arista.com
protocol instance agni-server
!

ssl profile agni-server
    certificate self-signed.crt key self-signed.key
    trust certificate AGNI_https_server_cert_chain.pem
!

ssl profile agni-server-radsec
    certificate auto-certificate myautocert
    trust certificate radsec_ca_certificate.pem
!

radius-server host radsec.beta.agni.arista.io tls ssl-profile agni-server-radsec
aaa group server radius agni-server-group
    server radsec.beta.agni.arista.io tls
!

aaa authentication dot1x default group agni-server-group
aaa accounting dot1x default start-stop group agni-server-group
!

dot1x system-auth-control
dot1x protocol lldp bypass
dot1x dynamic-authorization
dot1x
    radius av-pair service-type
    radius av-pair lldp system-name
    radius av-pair lldp system-description

```

Appendix/ Issues

1. <https://arista.my.site.com/AristaCommunity/s/article/Configuring-Arista-EST-Server>
2. <https://arista.my.site.com/AristaCommunity/s/article/Configuring-RadSec-profile-in-EOS>

Questions

Contact shaip@arista.com for any questions.